

Hybrid RAG for Financial 10-K Analysis (GAFAM)

Question answering grounded in SEC filings — hybrid text retrieval, with a roadmap for tabular querying (text-to-SQL) and intelligent routing.

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Corpus: 5 Form 10-Ks (Apple, Amazon, Alphabet, Meta, Microsoft)

Problem & objective

10-K reports mix dense narrative (risk factors, MD&A) with numerical financial statements. A standalone LLM hallucinates; pure keyword search misses paraphrases. This project builds a **RAG (Retrieval-Augmented Generation)** system that retrieves relevant passages and generates *cited* answers, with an architecture designed to extend toward SQL over financial tables.

Product architecture (3 versions)

V1 delivered Text RAG Full pipeline: PDF extraction → chunking → embeddings → Qdrant → hybrid search (dense + BM25, RRF fusion) → optional rerank → local generation (Ollama) → Streamlit UI → RAGAS evaluation.	V2 Tabular module Structured financial data in DuckDB ; numerical questions via text-to-SQL (no auto-routing yet). Complements text RAG where tables break under naive chunking.
V3 Orchestration Function calling: the system chooses text, SQL, or both, then merges answers. A single entry point for the analyst.	Value for recruiters NLP/LLMs applied to finance, modern retrieval, evaluation, a working demo, and a clear vision spanning unstructured ↔ structured data.

V1 — technical pipeline (delivered)

1. Extraction Docling: 10-K PDFs → structured Markdown (headings, blocks).	2. Chunking Hybrid: split on Markdown headings, then max size + overlap.	3. Index BGE-small embeddings (384-d); local Qdrant (cosine) + company/section metadata.	4. Answer Hybrid RRF ± MiniLM rerank; Ollama qwen2.5:3b; Streamlit demo; RAGAS.
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Key design choices: hybrid retrieval to fix semantic false friends (e.g. “supply agreements” vs true supply-chain risks); local generation for zero API cost; company filters to reduce cross-GAFAM noise. **Indicative V1 scale:** ~2,200 chunks indexed across 5 filings.

Stack & deliverables

Stack: Python · Docling · sentence-transformers (BGE) · Qdrant · rank-bm25 · Ollama · Streamlit · RAGAS · (optional Claude API) · DuckDB
planned in V2 | **Deliverables:** modular code (extraction, chunking, embeddings, indexing, retrieval, generation, evaluation) · demo UI · eval dataset · one-pager

Acknowledged limits & learnings

- Financial tables remain fragile in text-only V1 → explicitly motivate SQL V2.
- Lightweight MiniLM reranker can underperform hybrid RRF alone — hence measure with RAGAS.
- Local RAGAS judge (small LLM): indicative scores for iteration, not lab-grade benchmarks.